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Professor Bruce C. Berndt
Editor-in-Chief, *The Ramanujan Journal*
Springer

Dear Professor Berndt,

I am pleased to submit "A bounded two-tier null result for integer relations on a hybrid Khinchin-$K_0$ basis, with a documented capability gap and a Lean 4 statement-shape encoding" for consideration in *The Ramanujan Journal*.

The manuscript reports a cascade-stable PSLQ null on a 15-dimensional hybrid basis built around Khinchin's constant $K_0$, presented under a verification-class taxonomy that distinguishes a field-standard-practice empirical-tier bound ($H^{\mathrm{op}}_{\mathrm{emp}} \approx 7.997 \times 10^{69}$) from a rigorous Ferguson–Bailey–Arno (1999) proven-corollary bound ($H_{\mathrm{rig}} = 1.0361 \times 10^{72}$). Three Excluded Families precisely delineate the bounded scope from neighbouring results in the literature. A symbolic-closure attempt across seven candidate structural arguments returned a documented fundamental capability gap tied to $K_0$'s unresolved transcendence status, and the result statement is additionally encoded in Lean 4 against Mathlib4 v4.14.0 with an explicit auxiliary-axiom trust boundary.

The contribution sits squarely in the Bailey–Borwein–Plouffe experimental-number-theory tradition that *The Ramanujan Journal* regularly publishes, and addresses a question — integer relations on a Khinchin-mixed basis — that is directly relevant to the journal's scope on constants whose arithmetic nature remains open. The two-tier verification-class taxonomy and the documented capability-gap framing are intended to be reusable by other experimental investigations of similarly transcendence-open constants.

All research artefacts — the manuscript source, the cascading-precision PSLQ sweep outputs for the primary basis and the 65 scanned sub-bases, the environment lock file, the literature claims ledger, and the Lean 4 encoding — are deposited at Zenodo under concept DOI 10.5281/zenodo.20246707 (version DOI 10.5281/zenodo.20246708, CC-BY-4.0), with the reproducibility-pinned source repository at github.com/papanokechi/khinchin-k0-bounded (tag `gold/M6`).

The manuscript has not been published previously and is not under consideration at any other journal. I am an independent researcher with no institutional affiliation; The Ramanujan Journal's explicit support for independent authors was a material factor in venue selection. I have used AI-assisted execution as a research tool under an explicit accountability discipline disclosed in the manuscript's Statements and Declarations section, and I retain full accountability for all claims, methodology choices, and the final result.

Thank you for your consideration.

Sincerely,

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